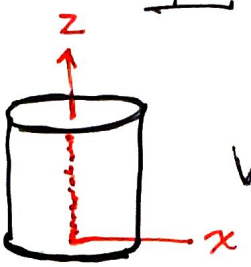
# Moment of Inertia

$$I = \int r^2 \, dm \longrightarrow r^2 \int dm$$




$$V = \pi r^2 \cdot h \longrightarrow \pi z^2(x) \cdot x$$

$$\frac{dV}{dx} = \pi \left( 2 z(x) \cdot \dot{z}(x) \cdot x + z^2(x) \right)$$

prove: $\dfrac{d z(x)}{dx} \cdot x = z(x)$

Simple proof; $z(x) \dfrac{dx}{dx} = z(x)$

$2^{nd}$ proof; $d z(x) \cdot x = z(x) \, dx$

$$\frac{1}{z(x)} dz(x) = \frac{1}{x} dx$$

$h(z(x)) = h(x)$ ; $c_{1,2} = 0$

$z(x) = x \longrightarrow \pi \left( 2 z(x) \cdot \dot{z}(x) \cdot x + z^2(x) \right)$

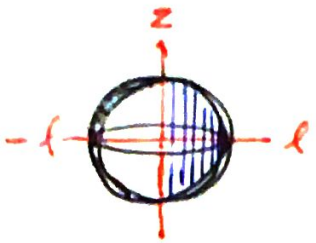
True!

$$\pi \left( 2 z(x) \cdot (1) \cdot z(x) + z^2(x) \right)$$

Since $z(x) = x$ , $\dot{z}(x) = 1$

Therefore $dV = 3 \pi z^2(x) \, dx$

# Inertia for the Sphere




$$dV = 3\pi z^2(x)\,dx$$

$$I_{rod} = \frac{1}{3}\ell^3 M \rightarrow \frac{1}{3}x^2\int \rho\, dV$$

"Integrating the mass 'rod by rod'"

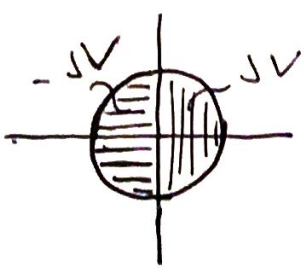
$$\frac{1}{3}x^2\int_{-\ell}^{\ell} 3\rho\pi z^2(x)\,dx \;;\; z(x) = \sqrt{\ell^2 - x^2}$$

$$\frac{1}{3}x^2\int_{-\ell}^{\ell} 3\rho\pi z^2(x)\,dx = x^2\int_{-\ell}^{\ell}\rho\pi(\ell^2 - x^2)\,dx$$

$$\rho\pi\int_{-\ell}^{\ell} x^2\ell^2 - x^4\,dx = \rho\pi\left(\left(\frac{\ell^5}{3} - \frac{\ell^5}{5}\right) - \left(\frac{(-\ell)^5}{3} - \frac{(-\ell)^5}{5}\right)\right)$$

$$= \rho\pi\left(\frac{4}{15}\ell^5\right) = \frac{3}{4}\frac{M}{\pi\ell^3}\frac{4}{15}\ell^5 = \frac{1}{5}M\ell^2$$

$V_{sphere} = \frac{4}{3}\pi\ell^3$




$$\Delta V = dV - (-dV) = 2\,dV$$

$$dV_{total} \rightarrow 2\cdot\frac{1}{5}M\ell^2$$

Integrating by Spherical Shells yields

$$I_{shell} = \frac{2}{3}m\ell^2$$

$$\rightarrow \frac{2}{3}x^2\int 3\rho\pi z^2(x)\,dx$$

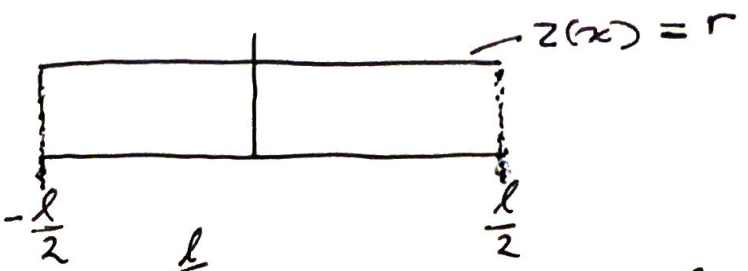
$$\rightarrow 2\,\frac{4}{15}\ell^5\rho\pi$$

$$\rightarrow 2\frac{4}{15}\ell^5\frac{3}{4}\frac{M}{\pi\ell^3}\cdot\pi = \frac{2}{5}M\ell^2$$

Derive the Rod about its center

using $dV = 3\pi z^2(x)\,dx$

$$\frac{1}{3}x^2 \int 3\pi\rho z^2(x)\,dx \;;\; z(x) = r$$




$$\rho\pi r^2 \int_{-\frac{\ell}{2}}^{\frac{\ell}{2}} x^2\,dx = \rho\pi r^2 \frac{x^3}{3}\Big|_{-\frac{\ell}{2}}^{\frac{\ell}{2}}$$

$$\rho\pi r^2\left(\frac{\ell^3}{8}\cdot\frac{1}{3} + \frac{\ell^3}{8}\cdot\frac{1}{3}\right)$$

$$\rho\pi r^2\left(\frac{2}{24}\ell^3\right) = \pi\rho r^2 \frac{1}{12}\ell^3$$

Volume of a rod $= \pi r^2 \cdot \ell$

$$\pi\frac{M}{\pi\cdot r^2\cdot\ell}\,\frac{1}{12}\ell^3 = \frac{1}{12}M\ell^2$$

Inertia of a rod about its center

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$$\pi\rho r^2 \frac{x^3}{3}\Big|_0^{\ell} \longrightarrow \pi\frac{M}{\pi r^2\ell} r^2\ell^3$$

$$= \frac{1}{3}M\ell^2$$

Inertia of a rod about its end.